MILITARY SPECIFICATION

# GAGE, SNAP, ADJUSTABLE, DIAL INDICATING

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1. Scope. This specification covers requirements for three types of Gage, Snap, Adjustable, Dial Indicating.

1.2 Classifications. Gage, Snap, Adjustable, Dial Indicating shall be of the following types, classes and sizes. (see 6.2)

1.2.1 Types, Classes and Sizes

Type I - Adjustable contact surface for wide areas with two fine adjustment features for zero setting.

Type II - Adjustable contact surface for narrow grooves with two fine adjustment features for zero setting.

Type III - Adjustable contact surface for wide areas with one fine adjustment feature for zero setting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMEDTA, Robins AFB, GA 31098-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5220

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NOTE: Further explanations of "Wide areas" and "narrow grooves" are given in respective paragraphs 3.4 and 3.5.

Class A - Indicator dial graduations in 0.0001 inches.

Class B - Indicator dial graduations in 0.0005 inches.

Sizes. Gage sizes are shown on Tables I, II, III and IV.

1.2.2 Specification Part Number. Military Specification shall include a specification based part numbering system to identify the parts or items covered. (see 3.12)

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and Standards. Unless otherwise specified the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specification and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

Federal

PPP-P-40 - Packaging and Packing of Hand Tools.

Military

MIL-I-18422 - Indicators, Dial and Accessories.

STANDARDS

Military

MIL-STD-130 - Identification Marking of US Military Property.
(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Design and Availability. The gage assembly shall be of the manufacturer's latest design that is commercially available and its design has been proven through usage. It shall have the capability of measuring between



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opposing parallel flat exterior surfaces and between opposing convex curved exterior surfaces within the applicable measurement range as specified in the applicable tables and showing measurement readings on a dial indicator. The gage assembly shall also have the capability of performing tests as specified in Section 4.

3.1.1 Handle. The gage assembly shall be equipped with a handle of a design and material that will assure a positive non-slip grip and will effectively isolate the gage assembly from body heat.

3.1.2 Backstop. The gage assembly shall be provided with a backstop that is adjustable to allow ease of diameter measurements between the minimum and maximum measurement range of the gage required.

3.1.3 Dial indicator guard. The gage assembly shall be provided with guards to protect the dial indicator in any attitude that the gage assembly can be in when it is laying on a flat surface.

3.1.4 Bench stand mounting provisions. The snap gage assembly shall be equipped with built-in provisions that would allow it to be mounted onto a bench stand.

3.1.5 Anvil retractor. When specified (see 6.2), types I and II gage assemblies shall be equipped with provisions for quickly retracting the sensitive measurement anvil for a distance of not less than 0.015 inches without interference with the zero setting.

3.1.6 Measurement surfaces. Unless otherwise specified (see 6.2), contacting surfaces of measurement anvils shall consist of tungsten carbide alloy. When specified (see 6.2), these contacting surfaces shall consist of steel alloy through hardened 55 to 65 on Rockwell "C" scale.

3.1.7 Zero setting adjustments. All provisions for zero setting of the gage assembly shall be built into the assembly without the need of using additional tools. These zero setting provisions shall be equipped with provisions for being latched in place, to prevent inadvertent movement.

3.2 Accuracy and repeatability. The gage assembly shall have accuracy and repeatability capability to the minimum requirement that within the Range 2-1/3 revolutions of the indicator, deviation from accurate measurement shall not exceed one dial graduation; and within the range in excess of 2-1/3 revolutions of the indicator, deviation from accurate measurement shall not exceed five dial graduations. When specified (see 6.2), proof of this accuracy and repeatability shall be made on each item to be delivered.

3.2.1 Dial indicator. The dial indicator used on the assembly shall have the accuracy and repeatability as required in Para 3.2 (for the gage assembly) and shall comply with all requirements of the current issue of MIL-I-18422 with its associated QPL.



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3.3 Bench stand. When specified (see 6.2), the snap gage assembly shall be accompanied by a bench stand. While the bench stand is setting on a horizontal flat surface, it shall be capable to securely holding the snap gage in an upright position for measuring between horizontal surfaces.

3.4 Type I gages. Type I gages shall be equipped with two opposing flat contacting measuring surfaces that will retain parallelism between measuring surfaces within 0.000150 of an inch for any adjustment setting of the gage assembly. Type I gages shall be capable of making measurements between maximum and minimum size measurements as specified in Table I for the gage size specified (see 6.2). Type I gages shall have a fine adjustment for moving the contacting surfaces, in relation to each other, for obtaining zero adjustment, and shall also allow the bezel on the indicator to be adjustable for zero setting.

TABLE I - RANGE SIZES APPLICABLE TO TYPE I

| SIZE | MEASUREMENT RANGE (INCHES) Minimum | Maximum |
|---|---|---|
| A | 0 | 1 |
| B | 1 | 2 |
| C | 2 | 3 |
| D | 3 | 4 |
| E | 4 | 5 |
| F | 5 | 6 |
| G | 6 | 7 |
| H | 7 | 8 |
| I | 8 | 9 |
| J | 9 | 10 |
| K | 10 | 11 |
| L | 11 | 12 |
| M | 12 | 13 |
| N | 13 | 14 |

3.5 Type II gages. Type II gages shall be equipped with two opposing, nonremovable, flat contacting measuring surfaces that will retain parallelism between measuring surfaces within 0.000150 of an inch for any adjustment setting of the gage assembly. These opposing flat contacting measuring surfaces shall be capable of making measurements within and inclusive of range size requirement of Table II for the size specified (see 6.2), and shall be capable of measuring bottom diameters of narrow grooves between shoulders no less than 3/4 inch apart. Type II gages shall have a fine adjustment feature for moving the contacting surfaces, in relation to each other, for the adjustable for zero setting.

TABLE II - RANGE SIZES APPLICABLE TO TYPE II

| SIZE | MEASUREMENT RANGE (INCHES) Minimum | Maximum |
|---|---|---|
| P | 0 | 1 |
| Q | 1 | 2 |
| R | 2 | 3 |
| S | 3 | 4 |
| T | 4 | 5 |



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3.5.1 Groove measurement contacts. Type II gage assemblies shall be fitted with contacts as specified in Table III for measuring bottoms of grooves. The groove measuring contacts shall be of the sizes specified in Table III and shall be supplied as pairs (two of each size required) (see 6.2).

TABLE III - GROOVE MEASURING CONTACTS FOR TYPE II GAGES

| Size | GROOVE MEASUREMENT RANGES (INCHES) Maximum Groove Width | Maximum Groove Depth |
|---|---|---|
| AA | 0.010 | 0.030 |
| AB | 0.027 | 0.040 |
| AC | 0.044 | 0.198 |
| AD | 0.084 | 0.250 |
| AE | Not Required | |

3.6 Type III gages. Type III gages shall be equipped with two opposing contacting measuring surfaces, of which one surface shall be flat and the other surface shall have a convex curvature. On the Type III gage assembly, fine adjustment for zero setting shall be accomplished by movement of the bezel on the indicator. Measurement range sized of Type III gages shall be specified (see 6.2) as applicable to Table IV.

TABLE IV - RANGE SIZES APPLICABLE TO TYPE III

| Size | MEASURING RANGE (INCHES) Minimum | Maximum |
|---|---|---|
| BA | 0 | 1 |
| BB | 1 | 2 |
| BC | 2 | 3 |
| BD | 3 | 4 |
| BE | 4 | 6 |
| BF | 6 | 8 |
| BG | 8 | 10 |
| BH | 10 | 12 |
| BJ | 12 | 14 |

3.7 Operation and Calibration Tools. Any tools required for normal shop usage for use in calibration shall accompany each gage assembly required.

3.8 Case. The gage assembly and its components shall be arranged in a case of the manufacturer's design in a manner that will give protection from damage in shipment and normal shop use.

3.9 Painting. Unless otherwise specified (see 6.2), the gage shall be painted or finished in accordance with the manufacturer's standard commercial practive.

3.10 Nameplate. Unless otherwise specified (see 6.2), a nameplate (decal, ink stamped, embossed or equal) shall be attached to the case of each gage assembly. The nameplate shall conform to MIL-STD-130 and contain the following data: Nomenclature, Manufacturer's name or identification, Manufacturer's model number, National Stock Number, U.S.



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3.11 Reclaimed materials. The assemblies shall contain reclaimed materials to the maximum extent possible without jeopardizing the material quality or performance of the equipment. The reclaimed material shall be reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical compositions and physical properties as the materials originally selected for use on the assemblies.

3.12 Definitive Specification Part Numbers. The specification part number is a definitive part which will be formulated to identify each item or part covered by this specification. The part number will be formulated by selecting from the requirement options available in this specification as follows:

Definitive Specification Part Number - M87955 X X XX XX X

Military Specification Number
Type Designator (see Table V and 3.12.1)
Class Designator (see Table VI)
Range Size Designator (see Table I, II, IV)
Groove Measurement Ranges (see Table III)
Anvil Retractor Designator (see 3.12.5, 6.2C)
Bench Stand (see 3.12.6, 6.2F, Table VIII)

3.12.1 Type Designator. A one position field shall be used to designate the required type of gage.

TABLE V

| Part Number Designator | Specification Type | Remarks |
|---|---|---|
| 1 | I | See 3.4 |
| 2 | II | See 3.5 |
| 3 | III | See 3.6 |

3.12.2 Class Designator. A one position field shall be used to designate indicator graduations.

TABLE VI

| Part NR Designator | |
|---|---|
| 5 | Class A; Indicator dial graduations in 0.0001 inches |
| 6 | Class B; Indicator dial graduations in 0.0005 inches |



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3.12.3 Range Sizes Designator. A one or two position field shall be used to designate the required measurement range. (see paragraphs 3.4, 3.5 for Type I, II, and paragraph 3.6 for Type III gages.)

3.12.4 Groove Measurement Designator. A two position field shall be used to designate the required measuring contacts. (see 3.5.1)

3.12.5 Anvil Retractor Designator. A one position field shall be used to designate whether the item is required.

TABLE VII

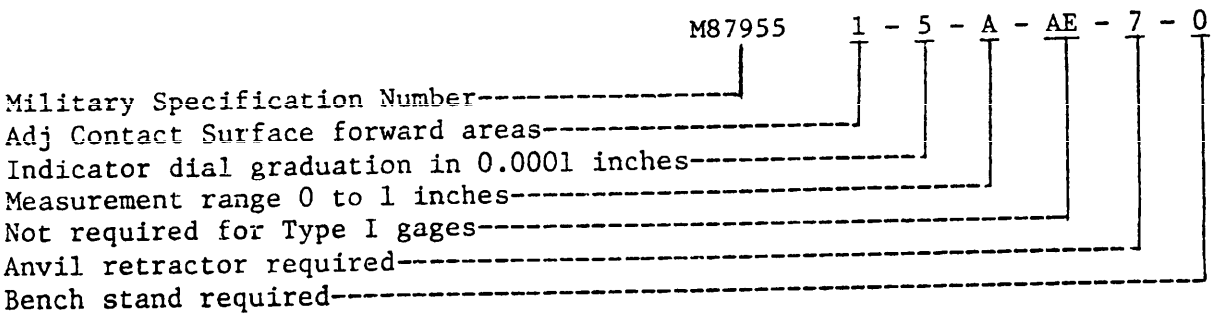
| Part Number Designator | Remarks (for Type I and II only) |
|---|---|
| 7 | Anvil Retractor Required |
| 8 | Anvil Retractor Not Required |

3.12.6 Bench Stand Designator. A one position field shall be used to designate whether the item is required.

TABLE VIII

| Part Number Designator | Remarks |
|---|---|
| 9 | Bench stand is required |
| 0 | Bench stand is not required |

TYPE I SAMPLE PART NUMBERS




4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.2 Preproduction inspection. Due to the requirement of this specification for commercially available items, there is no requirement for first article or preproduction inspection.

4.3 Quality conformance inspection. Each gage assembly offered for delivery under this specification shall be subjected to accuracy and repeatability testing. These tests shall consist of making at least three (3) different measurements of each of three (3) different sized diameters or thicknesses of objects within the measurement range of the gage assembly. This accuracy and repeatability testing shall be in compliance with requirements of 3.2. The objects that are to be used for making measurement tests shall be selected by the producer and unless otherwise specified (see 6.2), delivery of these objects will not be required.

4.4 Visual inspection. One gage assembly that is identical to the other gage assemblies offered, shall be visually inspected for any feature that would prevent it from making measurements of convex rounded surfaces and for all other detailed requirements as specified herein.

4.5 Rejection and retest. Any deviation from above requirements shall be cause for rejection until corrective actions are taken on all items offered and testing requirements (as applicable) be repeated.

4.6 Inspection of preparation for delivery. Preservation, packaging, packing, and marking for shipment and storage shall be inspected to determin conformance to the requirements of Section 5 or to solicitation requirements.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified in the solicitation requirements, preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The levels of preservation, packaging and packing shall be as specified (see 6.2 and solicitation requirements).

## 6. NOTES

6.1 Intended use. The gages covered in this specification are intended for use in repair shops, i.e., engine overhaul, machine, inspection, etc. They are essentially for use in determining shaft, bearing, and bushing sizes, but may also determine even lobed out-of-round and tapered conditions.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification along with identification and date of any amendment thereto.

b. Type, class and size of gage assembly required (see 1.2, 3.4, 3.5, 3.6, Tables I, II and IV).

c. Anvil retractor, if required, for Types I and II gages (see 3.1.5).

d. Contacting measurement surfaces, if other than tungsten carbide alloy (see 3.1.6).



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e. Report on accuracy and repeatability tests, if required (see 3.2).

f. Bench stand, if required (see 3.3).

g. Sizes of groove measuring contacts required for Type II gage (see 3.5.1 and Table III).

h. Finish, if different (see 3.9).

i. Nameplate, if different (see 3.10).

j. Delivery of objects used in accuracy and repeatability testing, if required (see 4.3).

k. Levels of preservation, packing, packaging, and marking required (see Section 5 and solicitation requirements).

Custodian:

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Review Activities:

70, 71, 80, 82

Users:

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